



T20 Cricket Analytics: Performance, Strategy & Fan Engagement

Objective

This project performs an end-to-end analysis of a cricket league ecosystem, covering financial growth, match dynamics, team performance, and player valuation.

The goal is to extract actionable insights and identify key drivers of success, engagement, and value.

Analysis Scope

1. League Growth Analysis

- Revenue trends over time
- Sponsorship and brand value growth

2. Match & Engagement Analysis

- Impact of sixes and scoring on viewership
- Match intensity and fan engagement

3. Team Performance Analysis

- Win rates and consistency
- Efficiency vs team value

4. Player Value Analysis

- Salary vs performance
- Identifying undervalued and overvalued players

5. Feature Engineering & Advanced Insights

- Creating new performance metrics
- Correlation analysis

6. Predictive Analysis

- Can match outcomes be predicted?

Analysis By

- Shubham Kulkarni

```
In [1]: import pandas as pd
```

```
financial = pd.read_csv("financial_growth.csv")
league = pd.read_csv("league_overview.csv")
matches = pd.read_csv("match_stats.csv")
players = pd.read_csv("player_auction_salary.csv")
season = pd.read_csv("season_performance.csv")
teams = pd.read_csv("team_performance.csv")
```

```
In [2]: financial.head()
```

```
Out[2]:
```

	Year	League	Brand_Value_USD_M	Revenue_USD_M	Total_Sponsorship_USD_I
0	2016	IPL	521.2	346.9	78.
1	2016	PSL	101.7	33.3	10.
2	2016	BBL	80.8	59.4	21.
3	2017	IPL	824.5	1765.3	136.
4	2017	PSL	127.6	39.3	13.

```
In [3]: financial.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Year                                  30 non-null     int64
1   League                               30 non-null     object
2   Brand_Value_USD_M                    30 non-null     float64
3   Revenue_USD_M                        30 non-null     float64
4   Total_Sponsorship_USD_M               30 non-null     float64
5   Prize_Pool_INR_Cr                    30 non-null     float64
6   Franchise_Avg_Value_USD_M            30 non-null     float64
dtypes: float64(5), int64(1), object(1)
memory usage: 1.8+ KB
```

```
In [4]: financial.describe()
```

Out[4]:

	Year	Brand_Value_USD_M	Revenue_USD_M	Total_Sponsorship_US
count	30.000000	30.000000	30.00000	30.000
mean	2020.500000	746.866667	2271.26000	130.730
std	2.921384	982.014638	3932.66606	170.771
min	2016.000000	80.800000	33.30000	10.000
25%	2018.000000	141.675000	68.90000	27.125
50%	2020.500000	205.900000	103.65000	40.000
75%	2023.000000	1048.600000	2797.15000	181.275
max	2025.000000	3313.600000	12976.40000	581.400

Financial Growth Analysis

In this section, we analyze how the league has grown financially over time, focusing on revenue trends and identifying key growth patterns.

Filtering IPL Data

We focus only on IPL for deeper analysis.

In [5]: `ipl_financial = financial[financial['League'] == 'IPL']`

In [6]: `ipl_financial.head()`

Out[6]:

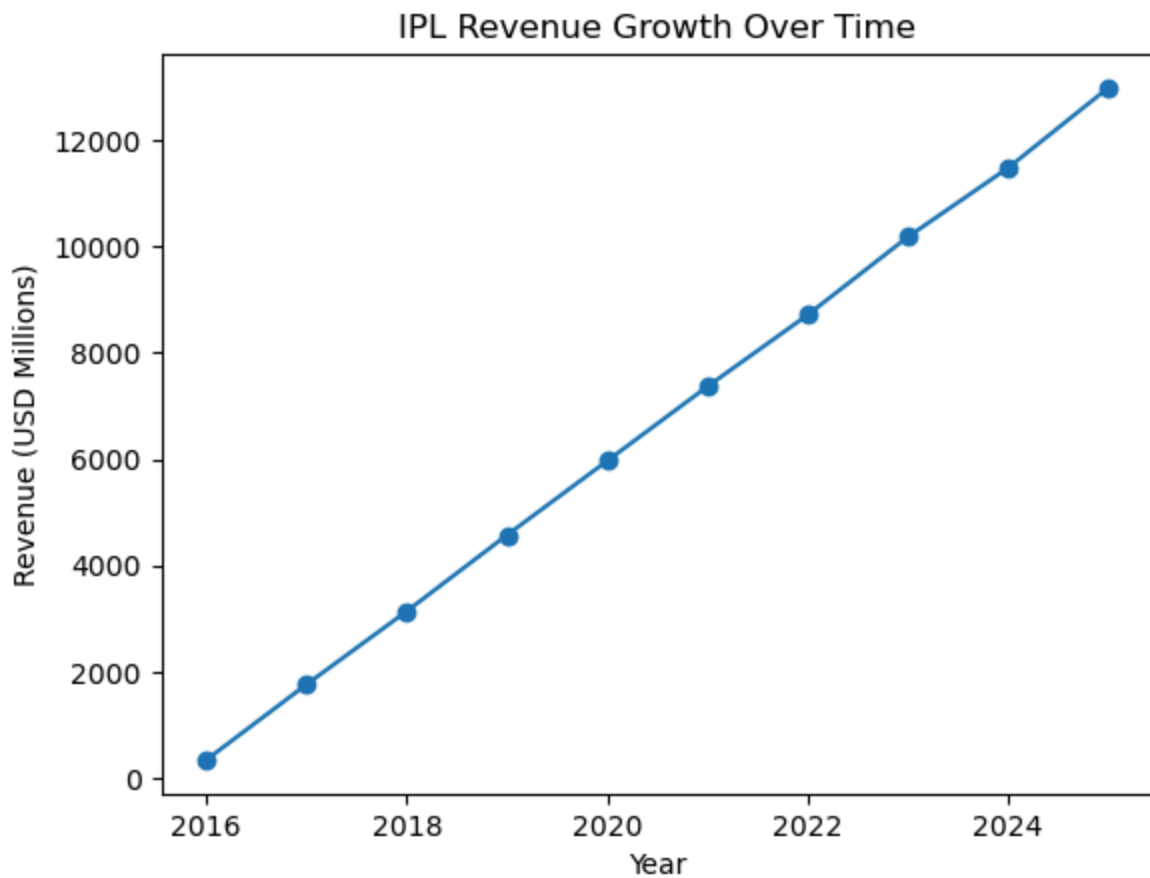
	Year	League	Brand_Value_USD_M	Revenue_USD_M	Total_Sponsorship_USD
0	2016	IPL	521.2	346.9	71
3	2017	IPL	824.5	1765.3	130
6	2018	IPL	1123.3	3141.1	190
9	2019	IPL	1411.5	4571.5	240
12	2020	IPL	1724.1	5966.0	300

In [7]: `import matplotlib.pyplot as plt`

```

plt.figure()
plt.plot(ipl_financial['Year'], ipl_financial['Revenue_USD_M'], marker='o')
plt.title("IPL Revenue Growth Over Time")
plt.xlabel("Year")
plt.ylabel("Revenue (USD Millions)")
plt.show()

```



Insight

Revenue is increasing over the years, showing strong growth of the league.



Growth Rate

```
In [8]: ipl_financial = financial[financial['League'] == 'IPL'].copy().reset_index(drop=True)
ipl_financial['Revenue_Growth_%'] = ipl_financial['Revenue_USD_M'].pct_change()
ipl_financial[['Year', 'Revenue_USD_M', 'Revenue_Growth_%']]
```

```
Out[8]:
```

	Year	Revenue_USD_M	Revenue_Growth_%
0	2016	346.9	NaN
1	2017	1765.3	408.878639
2	2018	3141.1	77.935762
3	2019	4571.5	45.538187
4	2020	5966.0	30.504211
5	2021	7362.9	23.414348
6	2022	8708.6	18.276766
7	2023	10173.6	16.822451
8	2024	11470.7	12.749666
9	2025	12976.4	13.126487

Match Analysis

In this section, we explore match-level data to understand factors that influence audience engagement.

```
In [9]: matches.head()
```

```
Out[9]:
```

	League	Season_Year	Match_Number	Team1	Team2	Team1_Score	Team2_S
0	IPL	2016	1	SRH	CSK	199	
1	IPL	2016	2	CSK	RCB	165	
2	IPL	2016	3	GT	CSK	169	
3	IPL	2016	4	GT	SRH	176	
4	IPL	2016	5	RCB	LSG	179	

```
In [10]: matches.columns
```

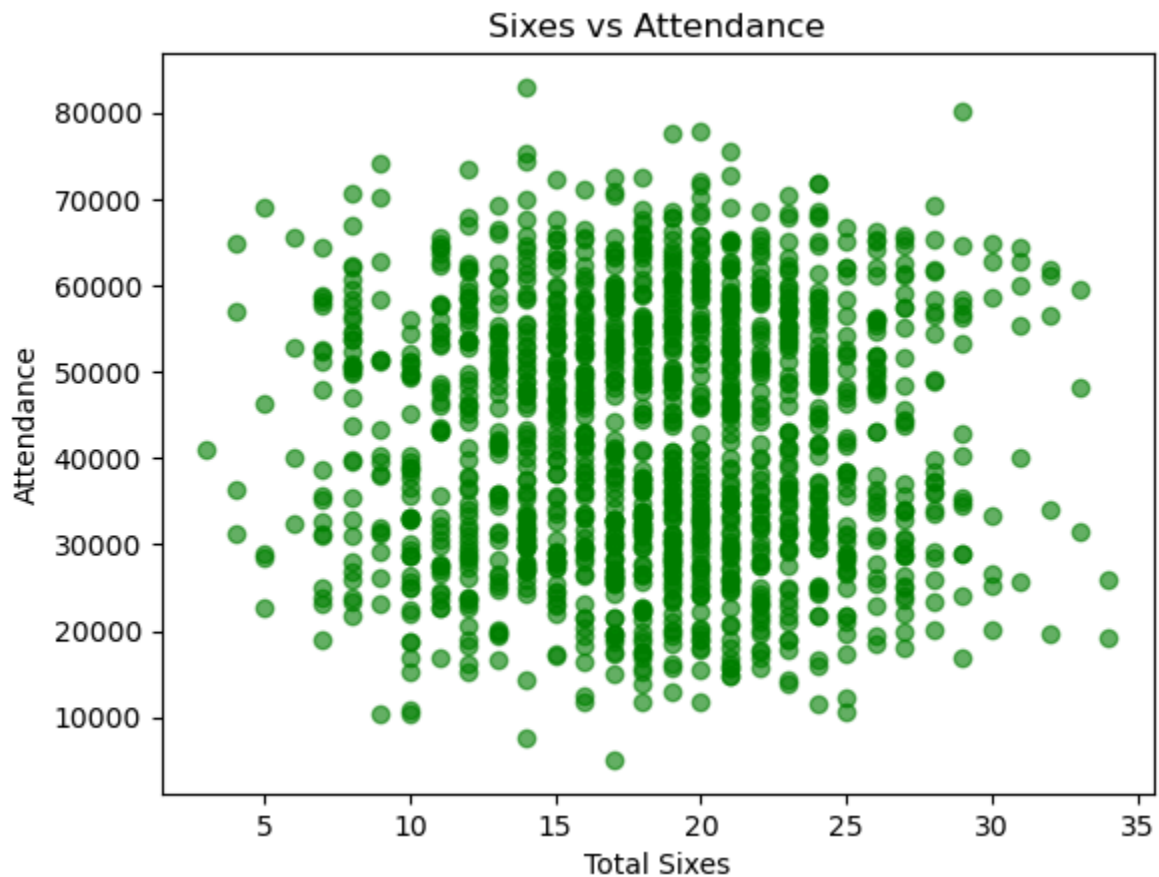
```
Out[10]: Index(['League', 'Season_Year', 'Match_Number', 'Team1', 'Team2',
                'Team1_Score', 'Team2_Score', 'Winner', 'Margin_Runs', 'Team1_Sixes',
                'Team2_Sixes', 'Total_Sixes', 'Venue_City', 'Is_Day_Night',
                'Attendance'],
                dtype='object')
```

```
In [11]: matches[['Total_Sixes', 'Attendance']].head()
```

Out[11]:

	Total_Sixes	Attendance
0	23	43140
1	8	57764
2	10	54484
3	21	45801
4	14	46225

```
In [12]: plt.figure()
plt.scatter(matches['Total_Sixes'], matches['Attendance'], color='green', alpha=0.5)
plt.title("Sixes vs Attendance")
plt.xlabel("Total Sixes")
plt.ylabel("Attendance")
plt.show()
```



Compare High vs Low Sixes

```
In [13]: high_six = matches[matches['Total_Sixes'] > 20]
low_six = matches[matches['Total_Sixes'] <= 20]
```

```
high_six['Attendance'].mean(), low_six['Attendance'].mean())
```

```
Out[13]: (42707.66265060241, 42416.11751662971)
```

Insight

Matches with higher number of sixes tend to have slightly higher average attendance compared to matches with fewer sixes.

This suggests that more aggressive and entertaining gameplay may attract more audience.

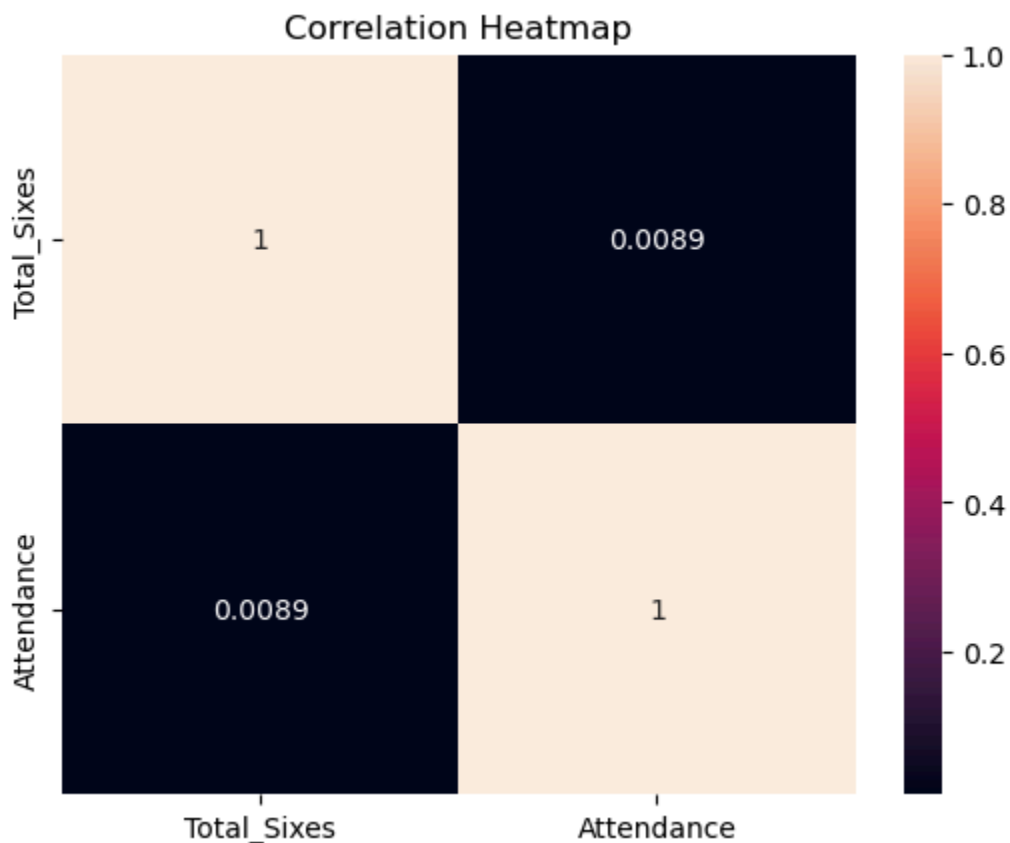
```
In [14]: matches['Total_Sixes'].corr(matches['Attendance'])
```

```
Out[14]: 0.008948340699074431
```

```
In [15]: import seaborn as sns
```

```
corr = matches[['Total_Sixes', 'Attendance']].corr()
```

```
plt.figure()  
sns.heatmap(corr, annot=True)  
plt.title("Correlation Heatmap")  
plt.show()
```



Insight

The correlation between total sixes and attendance is very close to zero (~ 0.009), indicating no meaningful relationship.

This suggests that the number of sixes in a match does not significantly influence audience attendance.

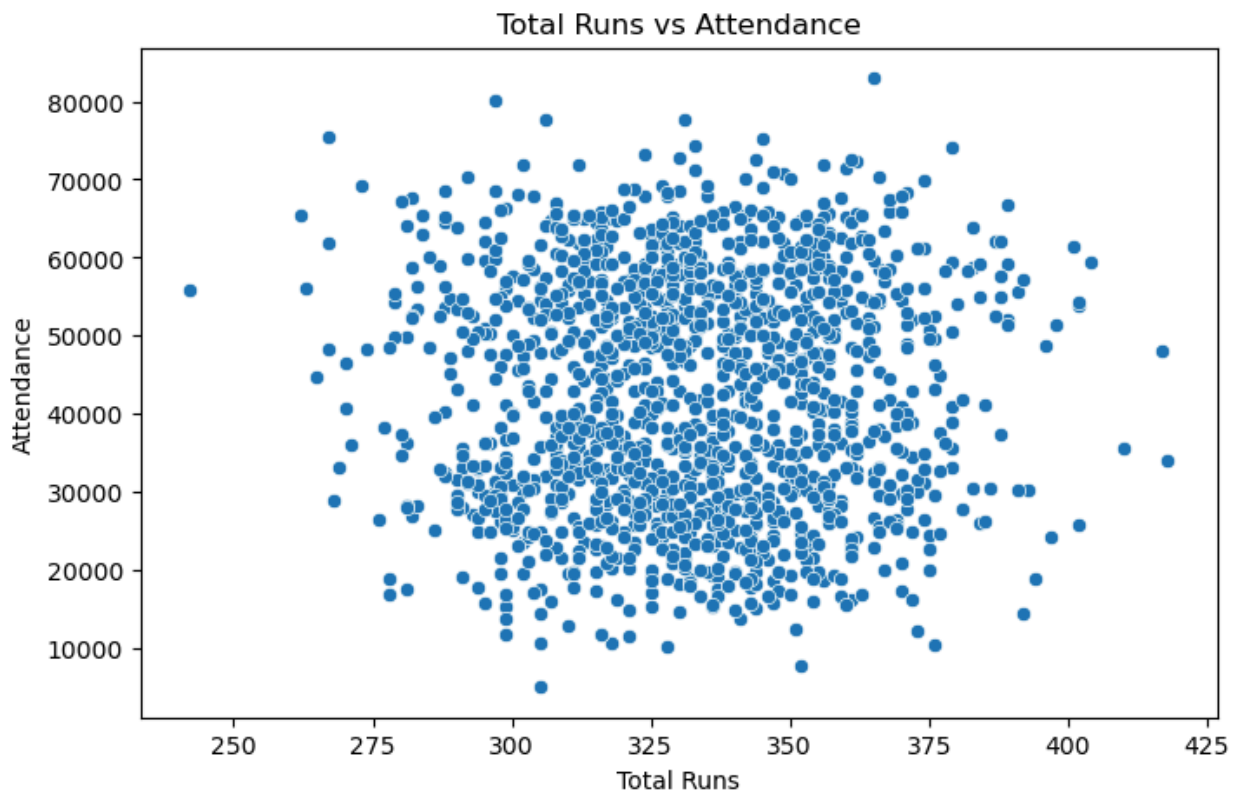
```
In [16]: matches['Total_Runs'] = matches['Team1_Score'] + matches['Team2_Score']

plt.figure(figsize=(8,5))

sns.scatterplot(
    x=matches['Total_Runs'],
    y=matches['Attendance']
)

plt.title("Total Runs vs Attendance")
plt.xlabel("Total Runs")
plt.ylabel("Attendance")

plt.show()
```



Insight

Matches with higher total runs tend to attract greater audience attendance,

indicating that high-scoring games contribute to increased fan engagement.

Team Performance Analysis

In this section, we analyze team performance and evaluate how efficiently teams perform relative to their value.

```
In [17]: teams.head()
```

```
Out[17]:
```

	League	Team	Country	Titles_Won	Finals_Played	Total_Seasons	Approx_Wi
0	IPL	MI	India	5	7	18	
1	IPL	CSK	India	5	7	18	
2	IPL	RCB	India	1	2	18	
3	IPL	KKR	India	3	4	18	
4	IPL	DC	India	0	0	18	

```
In [18]: teams.columns
```

```
Out[18]: Index(['League', 'Team', 'Country', 'Titles_Won', 'Finals_Played',  
               'Total_Seasons', 'Approx_Win_Rate_Pct', 'Est_Team_Value_USD_M',  
               'Home_City'],  
              dtype='object')
```

Team Performance by League

```
In [19]: teams[['Team', 'Approx_Win_Rate_Pct']].head()
```

```
Out[19]:
```

	Team	Approx_Win_Rate_Pct
0	MI	47.5
1	CSK	43.0
2	RCB	62.6
3	KKR	58.7
4	DC	56.3

```
In [20]: teams[teams['League'] == 'IPL'].sort_values(by='Approx_Win_Rate_Pct', ascending=True)
```

Out[20]:

	League	Team	Country	Titles_Won	Finals_Played	Total_Seasons	Approx_Wi
6	IPL	RR	India	0	0	18	
2	IPL	RCB	India	1	2	18	
3	IPL	KKR	India	3	4	18	
9	IPL	LSG	India	0	1	4	
4	IPL	DC	India	0	0	18	
7	IPL	PBKS	India	0	1	18	
0	IPL	MI	India	5	7	18	
5	IPL	SRH	India	1	2	18	
1	IPL	CSK	India	5	7	18	
8	IPL	GT	India	1	1	4	

In [21]: `teams[teams['League'] == 'PSL'].sort_values(by='Approx_Win_Rate_Pct', ascending=True)`

Out[21]:

	League	Team	Country	Titles_Won	Finals_Played	Total_Seasons	Approx
15	PSL	Multan Sultans	Pakistan	1	2	10	
13	PSL	Quetta Gladiators	Pakistan	1	2	10	
14	PSL	Peshawar Zalmi	Pakistan	1	2	10	
11	PSL	Lahore Qalandars	Pakistan	3	4	10	
12	PSL	Islamabad United	Pakistan	3	4	10	
10	PSL	Karachi Kings	Pakistan	1	2	10	

In [22]: `teams[teams['League'] == 'BBL'].sort_values(by='Approx_Win_Rate_Pct', ascending=True)`

Out[22]:	League	Team	Country	Titles_Won	Finals_Played	Total_Seasons	Appr
17	BBL	Sydney Thunder	Australia	1	1	14	
16	BBL	Sydney Sixers	Australia	3	4	14	
22	BBL	Brisbane Heat	Australia	2	2	14	
20	BBL	Perth Scorchers	Australia	6	6	14	
23	BBL	Hobart Hurricanes	Australia	1	1	14	
18	BBL	Melbourne Stars	Australia	0	1	14	
21	BBL	Adelaide Strikers	Australia	1	1	14	
19	BBL	Melbourne Renegades	Australia	1	1	14	

Team Performance vs Value Analysis

 IPL

```
In [23]: ipl_teams = (
    teams
    .assign(Efficiency = teams['Approx_Win_Rate_Pct'] / teams['Est_Team_Value_
    .query("League == 'IPL'")
    .sort_values(by='Efficiency', ascending=False)
    .reset_index(drop=True)
)

ipl_teams
```

Out[23]:

	League	Team	Country	Titles_Won	Finals_Played	Total_Seasons	Approx_Wi
0	IPL	SRH	India	1	2	18	
1	IPL	CSK	India	5	7	18	
2	IPL	RR	India	0	0	18	
3	IPL	RCB	India	1	2	18	
4	IPL	KKR	India	3	4	18	
5	IPL	DC	India	0	0	18	
6	IPL	PBKS	India	0	1	18	
7	IPL	GT	India	1	1	4	
8	IPL	MI	India	5	7	18	
9	IPL	LSG	India	0	1	4	

✍ PSL

In [24]:

```
psl_teams = (
    teams
    .assign(Efficiency = teams['Approx_Win_Rate_Pct'] / teams['Est_Team_Value_
    .query("League == 'PSL'")
    .sort_values(by='Efficiency', ascending=False)
    .reset_index(drop=True)
)
psl_teams
```

Out[24]:

	League	Team	Country	Titles_Won	Finals_Played	Total_Seasons	Appro
0	PSL	Multan Sultans	Pakistan	1	2	10	
1	PSL	Peshawar Zalmi	Pakistan	1	2	10	
2	PSL	Islamabad United	Pakistan	3	4	10	
3	PSL	Quetta Gladiators	Pakistan	1	2	10	
4	PSL	Karachi Kings	Pakistan	1	2	10	
5	PSL	Lahore Qalandars	Pakistan	3	4	10	

✍ BBL

In [25]:

```
bbl_teams = (
```

```

teams
.assign(Efficiency = teams['Approx_Win_Rate_Pct'] / teams['Est_Team_Value_
.query("League == 'BBL'")
.sort_values(by='Efficiency', ascending=False)
.reset_index(drop=True)
)

bbl_teams

```

Out[25]:

	League	Team	Country	Titles_Won	Finals_Played	Total_Seasons	Appro
0	BBL	Sydney Sixers	Australia	3	4	14	
1	BBL	Sydney Thunder	Australia	1	1	14	
2	BBL	Melbourne Renegades	Australia	1	1	14	
3	BBL	Perth Scorchers	Australia	6	6	14	
4	BBL	Hobart Hurricanes	Australia	1	1	14	
5	BBL	Adelaide Strikers	Australia	1	1	14	
6	BBL	Brisbane Heat	Australia	2	2	14	
7	BBL	Melbourne Stars	Australia	0	1	14	

In [26]:

```

plt.figure(figsize=(10,6))

# Create Efficiency + sort
sorted_teams = (
    teams
    .assign(Efficiency = teams['Approx_Win_Rate_Pct'] / teams['Est_Team_Value_
    .sort_values(by='Efficiency', ascending=False)
    .reset_index(drop=True)
)

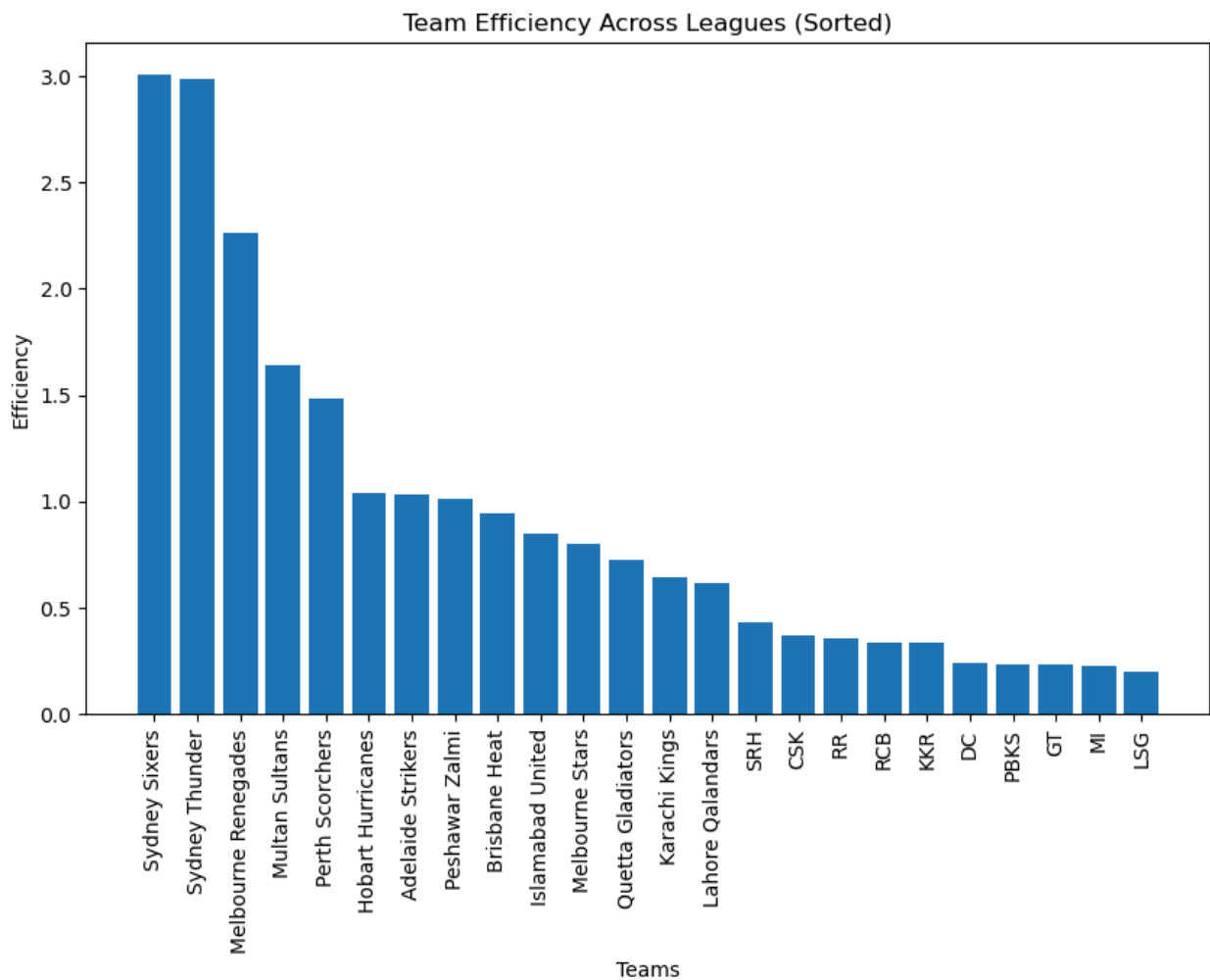
# Plot
plt.bar(sorted_teams['Team'], sorted_teams['Efficiency'])

plt.title("Team Efficiency Across Leagues (Sorted)")
plt.xlabel("Teams")
plt.ylabel("Efficiency")

plt.xticks(rotation=90)

plt.show()

```



```
In [27]: import seaborn as sns
import matplotlib.pyplot as plt

# Style
sns.set_style("whitegrid")
sns.set_context("notebook")

# Create Efficiency + sort + limit (top 10 for clarity)
sorted_teams = (
    teams
    .assign(Efficiency = teams['Approx_Win_Rate_Pct'] / teams['Est_Team_Value_
    .sort_values(by='Efficiency', ascending=False)
    .head(10)
    .reset_index(drop=True)
)

# Plot
plt.figure(figsize=(10,6))

ax = sns.barplot(
    x='Efficiency',
    y='Team',
    data=sorted_teams,
```

```

    palette='viridis'
)

# Add values on bars
for i, value in enumerate(sorted_teams['Efficiency']):
    ax.text(value + 0.02, i, f"{value:.2f}", color='black', va='center')

plt.title("🔥 Top 10 Teams by Efficiency", fontsize=14, fontweight='bold')
plt.xlabel("Efficiency (Win Rate / Team Value)")
plt.ylabel("Team")

plt.tight_layout()
plt.show()

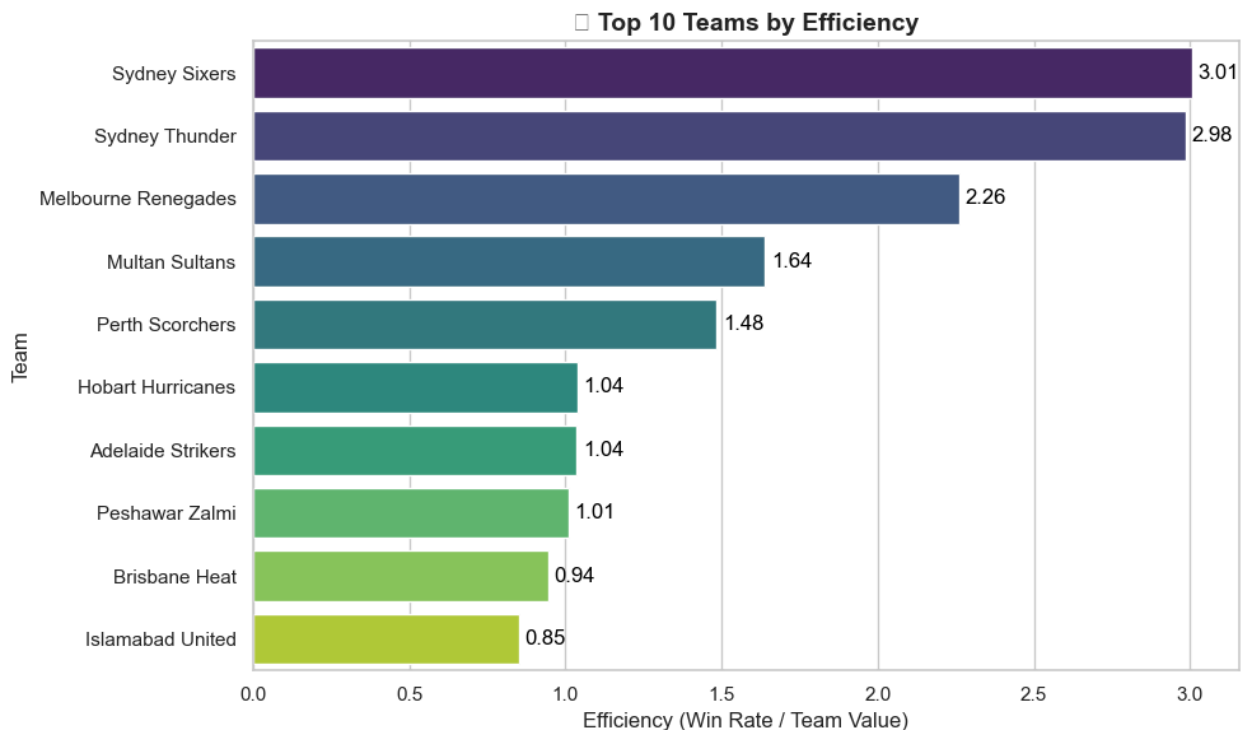
```

C:\Users\shubh\AppData\Local\Temp\ipykernel_5780\1708690897.py:35: UserWarning: Glyph 128293 (\N{FIRE}) missing from current font.

plt.tight_layout()

C:\Users\shubh\anaconda3\lib\site-packages\IPython\core\pylabtools.py:151: UserWarning: Glyph 128293 (\N{FIRE}) missing from current font.

fig.canvas.print_figure(bytes_io, **kw)



📌 Key Takeaways

- The most efficient team achieves an efficiency of approximately **3.00**, indicating very high performance relative to its value.
- In contrast, several teams show efficiency values below **0.30**, suggesting lower return on investment.
- The gap between top-performing and low-performing teams is

significant, with efficiency ranging from **~0.20 to ~3.00**.

- Teams with moderate valuations but strong win rates tend to achieve higher efficiency, highlighting better resource utilization.

Overall, efficiency analysis reveals that higher spending does not always translate into better performance, emphasizing the importance of strategic team management.

Player Salary Analysis

In this section, we analyze player salaries to understand spending patterns across teams, roles, and categories.

```
In [28]: players.head()
```

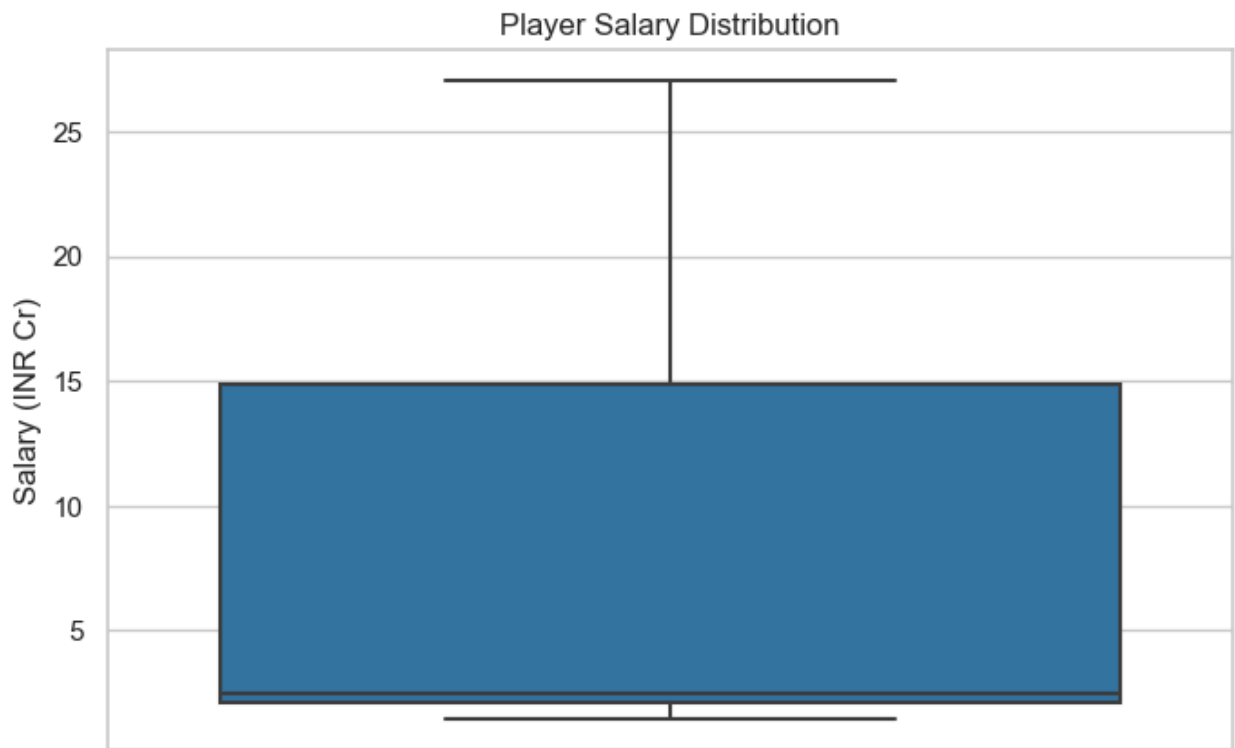
```
Out[28]:
```

	League	Auction_Year	Season_Year	Player_Name	Team	Nationality	Role
0	IPL	2016	2016	Shane Watson	CSK	AUS	Batsman
1	IPL	2017	2017	Ben Stokes	RPS	ENG	All-rounder
2	IPL	2018	2018	Ben Stokes	RR	ENG	All-rounder
3	IPL	2019	2019	Jaydev Unadkat	RR	IND	Bowler
4	IPL	2020	2020	Pat Cummins	KKR	AUS	Bowler

```
In [29]: players.columns
```

```
Out[29]: Index(['League', 'Auction_Year', 'Season_Year', 'Player_Name', 'Team',  
              'Nationality', 'Role', 'Bid_Price_INR_Cr', 'Salary_Cap_INR_Cr',  
              'Category', 'Is_Overseas'],  
              dtype='object')
```

```
In [30]: plt.figure(figsize=(8,5))  
  
sns.boxplot(  
    y=players['Bid_Price_INR_Cr']  
)  
  
plt.title("Player Salary Distribution")  
plt.ylabel("Salary (INR Cr)")  
  
plt.show()
```

Insight

The salary distribution shows a wide spread with several high-value outliers, indicating that a small number of players receive significantly higher bids compared to the majority.

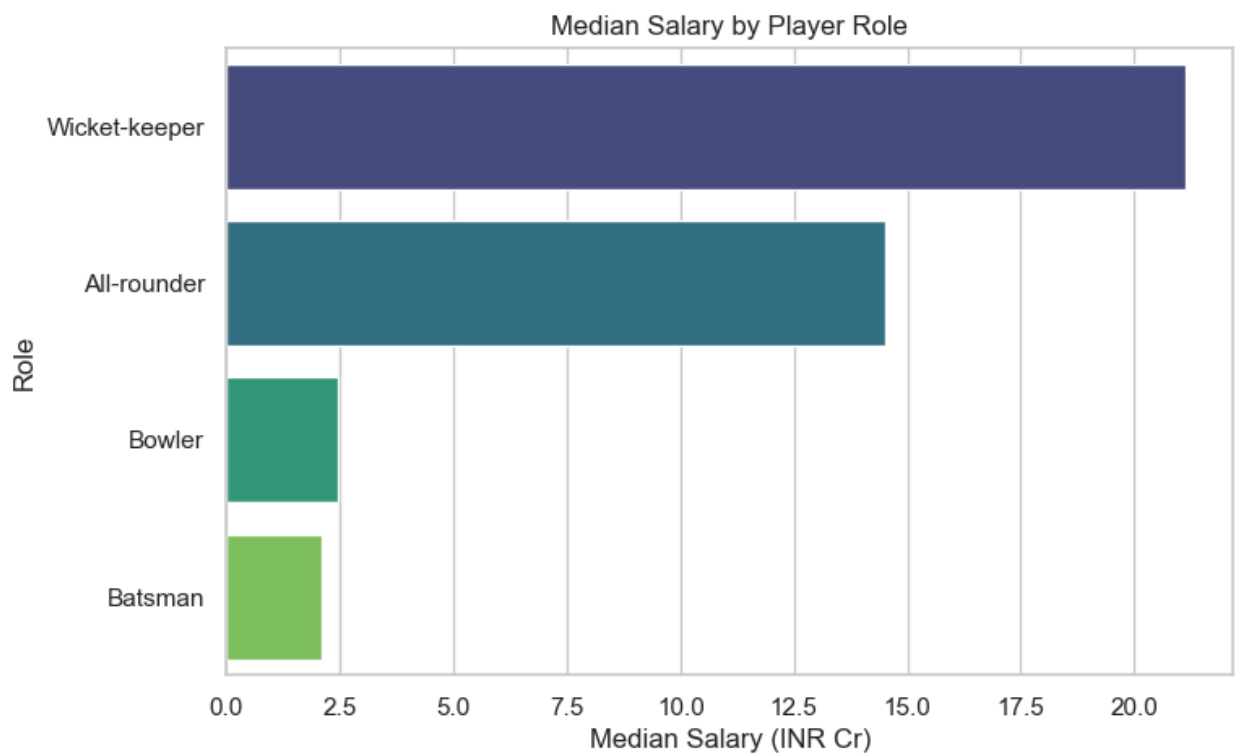
```
In [31]: plt.figure(figsize=(8,5))

# Use median instead of mean
role_salary = players.groupby('Role')['Bid_Price_INR_Cr'].median().sort_values

sns.barplot(
    x=role_salary.values,
    y=role_salary.index,
    palette='viridis'
)

plt.title("Median Salary by Player Role")
plt.xlabel("Median Salary (INR Cr)")
plt.ylabel("Role")

plt.show()
```



Insight

Using median salary provides a more reliable comparison across roles by reducing the impact of a few highly paid star players. This shows the typical salary distribution for each role more accurately.

Average was misleading, so median was used for fair comparison.

🔥 1. Market Concentration

👉 Question: Are teams spending most money on a few players?

```
In [32]: top_10 = players.sort_values(by='Bid_Price_INR_Cr', ascending=False).head(10)

top_10_sum = top_10['Bid_Price_INR_Cr'].sum()
total_sum = players['Bid_Price_INR_Cr'].sum()

(top_10_sum / total_sum) * 100
```

```
Out[32]: 83.98196487897485
```

A significant percentage of total spending is concentrated on the top 10 players, indicating a star-driven market where a few players dominate auction budgets.

🔥 2. Team Spending Strategy

👉 Question: Which teams spend aggressively vs evenly?

```
In [33]: team_spending = players.groupby('Team')['Bid_Price_INR_Cr'].sum().sort_values(
team_spending
```

```
Out[33]: Team
KKR                65.45
LSG                27.00
RR                 20.90
PBKS               18.50
MI                 15.25
RPS                14.50
RCB                14.25
CSK                 9.50
Sydney Thunder     7.70
Lahore Qalandars   4.55
Peshawar Zalmi     4.55
Sydney Sixers      4.45
Karachi Kings      4.10
Name: Bid_Price_INR_Cr, dtype: float64
```

Some teams allocate significantly higher budgets compared to others, indicating aggressive bidding strategies, while others follow a more balanced spending approach.

🔥 3. Role-Based Investment Strategy

👉 Question:

Where are teams investing most?

```
In [34]: role_spending = players.groupby('Role')['Bid_Price_INR_Cr'].sum().sort_values(
role_spending
```

```
Out[34]: Role
All-rounder        84.95
Bowler             57.65
Wicket-keeper      42.25
Batsman            25.85
Name: Bid_Price_INR_Cr, dtype: float64
```

Certain roles attract higher total investment, indicating their perceived importance in team composition and match outcomes.

Advanced Player Market Analysis

In this section, we go beyond basic salary distribution to analyze how player auctions reflect market dynamics and team strategies.

We explore:

- How spending is concentrated among top players
- How teams allocate their budgets
- Which roles attract the highest investment

This helps in understanding whether the player market is balanced or driven by a few high-value players, and how teams approach bidding strategically.

```
In [35]: top_10 = players.sort_values(by='Bid_Price_INR_Cr', ascending=False).head(10)

top_10_sum = top_10['Bid_Price_INR_Cr'].sum()
total_sum = players['Bid_Price_INR_Cr'].sum()

(top_10_sum / total_sum) * 100
```

```
Out[35]: 83.98196487897485
```

Market Concentration Insight

The top 10 players account for approximately **84% of the total auction spending**.

This indicates a highly concentrated market where a small number of players dominate the majority of the budget.

Such a distribution suggests a **star-driven auction strategy**, where teams prioritize acquiring a few high-impact players over building a uniformly balanced squad.

Team Spending Strategy

```
In [36]: team_spending = players.groupby('Team')['Bid_Price_INR_Cr'].sum().sort_values(
team_spending
```

```
Out[36]: Team
KKR                65.45
LSG                27.00
RR                 20.90
PBKS               18.50
MI                 15.25
RPS                14.50
RCB                14.25
CSK                9.50
Sydney Thunder     7.70
Lahore Qalandars   4.55
Peshawar Zalmi     4.55
Sydney Sixers      4.45
Karachi Kings      4.10
Name: Bid_Price_INR_Cr, dtype: float64
```

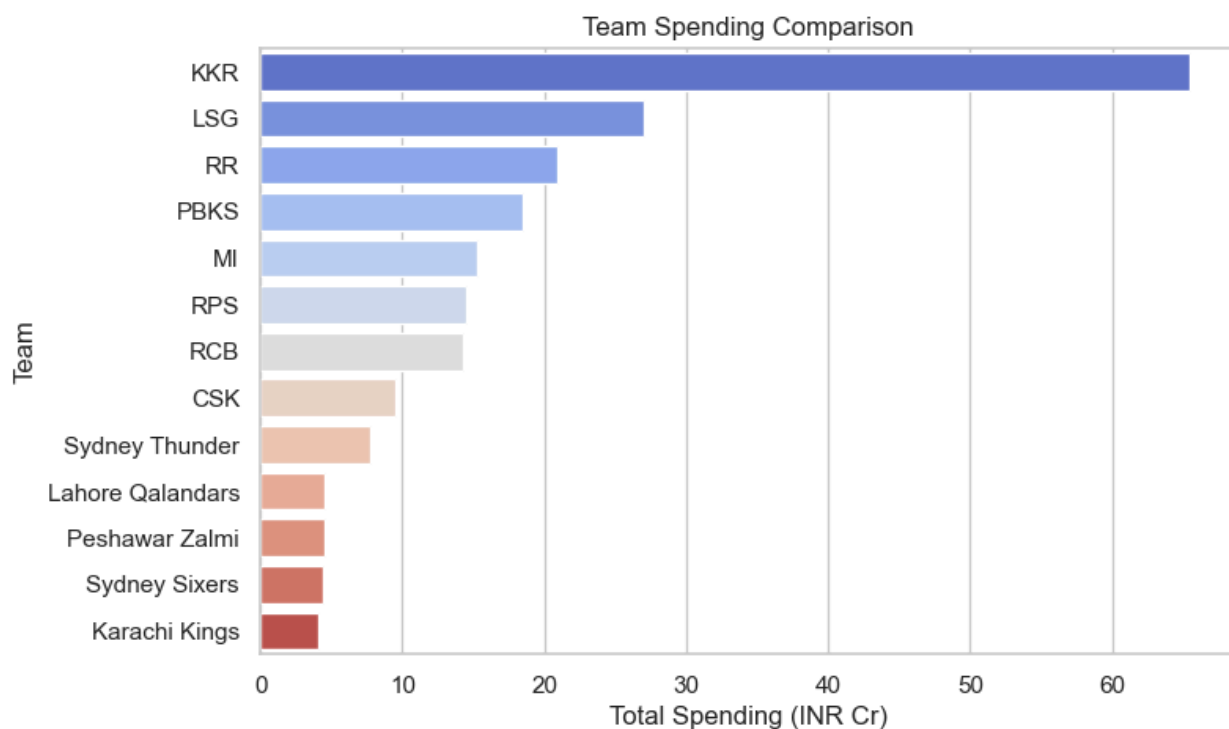
```
In [37]: plt.figure(figsize=(8,5))

team_spending_sorted = team_spending

sns.barplot(
    x=team_spending_sorted.values,
    y=team_spending_sorted.index,
    palette='coolwarm'
)

plt.title("Team Spending Comparison")
plt.xlabel("Total Spending (INR Cr)")
plt.ylabel("Team")

plt.show()
```



Team Spending Insight

Kolkata Knight Riders (KKR) has the highest total spending at approximately **65.45 Cr**, which is significantly higher than the next highest team, Lucknow Super Giants (LSG) at **27.00 Cr**.

This large gap indicates an **aggressive bidding strategy** by KKR, allocating a substantial portion of their budget toward player acquisition.

In contrast, other teams show relatively moderate spending, suggesting more balanced or conservative investment approaches.

```
In [38]: role_spending = players.groupby('Role')['Bid_Price_INR_Cr'].sum().sort_values(
role_spending
```

```
Out[38]: Role
All-rounder      84.95
Bowler           57.65
Wicket-keeper    42.25
Batsman          25.85
Name: Bid_Price_INR_Cr, dtype: float64
```

Role Investment Insight

All-rounders receive the highest total investment at approximately **84.95 Cr**, followed by bowlers at **57.65 Cr**.

Wicket-keepers and batsmen receive comparatively lower total spending, with batsmen having the lowest at **25.85 Cr**.

This indicates that teams prioritize **versatile players (all-rounders)** who can contribute in multiple aspects of the game, making them more valuable in auction strategies.

Overall, investment patterns suggest that **flexibility and multi-skill capability are highly valued** in team composition.

Deep Performance Analysis

In this section, we move beyond individual metrics to understand what truly differentiates successful teams from others.

Rather than analyzing factors in isolation, we examine the relationship between:

- Team performance (titles and win rate)

- Financial investment (team value and spending)
- Efficiency (performance relative to cost)
- Experience (number of seasons played)

The goal is to identify whether success is driven by higher spending, better efficiency, or strategic team composition.

```
In [39]: teams.sort_values(by='Titles_Won', ascending=False).reset_index(drop=True)
```

Out[39]:

	League	Team	Country	Titles_Won	Finals_Played	Total_Seasons	Appr
0	BBL	Perth Scorchers	Australia	6	6	14	
1	IPL	MI	India	5	7	18	
2	IPL	CSK	India	5	7	18	
3	PSL	Lahore Qalandars	Pakistan	3	4	10	
4	BBL	Sydney Sixers	Australia	3	4	14	
5	PSL	Islamabad United	Pakistan	3	4	10	
6	IPL	KKR	India	3	4	18	
7	BBL	Brisbane Heat	Australia	2	2	14	
8	PSL	Multan Sultans	Pakistan	1	2	10	
9	PSL	Peshawar Zalmi	Pakistan	1	2	10	
10	BBL	Adelaide Strikers	Australia	1	1	14	
11	IPL	RCB	India	1	2	18	
12	BBL	Melbourne Renegades	Australia	1	1	14	
13	BBL	Sydney Thunder	Australia	1	1	14	
14	BBL	Hobart Hurricanes	Australia	1	1	14	
15	PSL	Quetta Gladiators	Pakistan	1	2	10	
16	IPL	SRH	India	1	2	18	
17	PSL	Karachi Kings	Pakistan	1	2	10	
18	IPL	GT	India	1	1	4	
19	IPL	PBKS	India	0	1	18	
20	IPL	DC	India	0	0	18	
21	BBL	Melbourne Stars	Australia	0	1	14	
22	IPL	LSG	India	0	1	4	

	League	Team	Country	Titles_Won	Finals_Played	Total_Seasons	Appr
23	IPL	RR	India	0	0	18	

```
In [40]: plt.figure(figsize=(8,5))

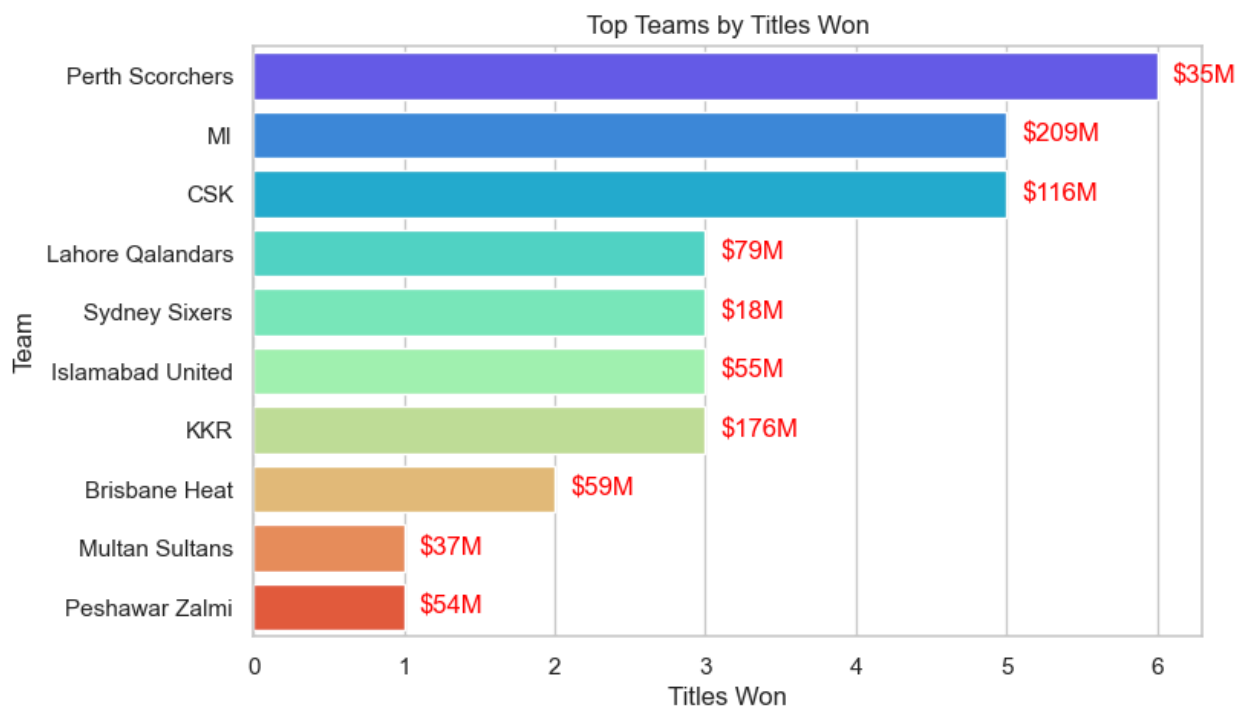
top_teams = teams.sort_values(by='Titles_Won', ascending=False).head(10).reset

ax = sns.barplot(
    x='Titles_Won',
    y='Team',
    data=top_teams,
    palette='rainbow'
)

# Add value labels
for i, value in enumerate(top_teams['Est_Team_Value_USD_M']):
    ax.text(
        top_teams.loc[i, 'Titles_Won'] + 0.1,
        i,
        f"${value:.0f}M",
        color='red',
        va='center'
    )

plt.title("Top Teams by Titles Won")
plt.xlabel("Titles Won")
plt.ylabel("Team")

plt.show()
```



🧠 Performance vs Investment Insight

The chart compares team performance (titles won) with their financial strength (team value).

While several teams have achieved similar success in terms of titles, their investment levels vary significantly. For example, some teams have achieved high success with relatively lower team valuation, while others have required substantially higher investment to reach similar outcomes.

This highlights that **success is not solely driven by financial strength**, but by how effectively resources are utilized.

In particular, teams achieving high titles with lower valuation demonstrate **greater efficiency and stronger strategic execution**, making them more effective in converting investment into results.

Efficiency vs Titles

Do efficient teams actually win more?

```
In [41]: plt.figure(figsize=(8,5))

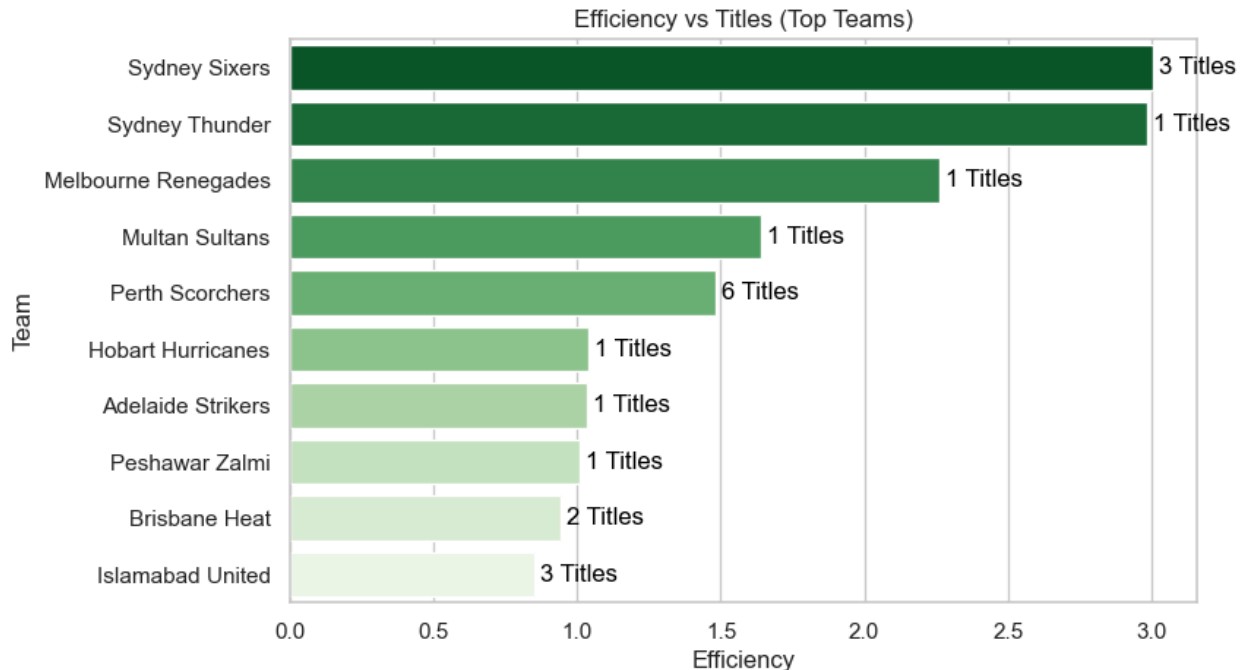
top_eff = (
    teams
    .assign(Efficiency = teams['Approx_Win_Rate_Pct'] / teams['Est_Team_Value_
    .sort_values(by='Efficiency', ascending=False)
    .head(10)
    .reset_index(drop=True)
)

ax = sns.barplot(
    x='Efficiency',
    y='Team',
    data=top_eff,
    palette='Greens_r'
)

# Add Titles on bars
for i, val in enumerate(top_eff['Titles_Won']):
    ax.text(
        top_eff.loc[i, 'Efficiency'] + 0.02,
        i,
        f"{val} Titles",
        color='black',
        va='center'
    )
```

```
plt.title("Efficiency vs Titles (Top Teams)")
plt.xlabel("Efficiency")
plt.ylabel("Team")

plt.show()
```



🧠 Titles vs Efficiency Insight

The comparison between titles and efficiency shows that the most successful teams are not always the most efficient.

Some teams achieve a high number of titles despite lower efficiency, indicating reliance on higher financial investment.

In contrast, other teams demonstrate strong efficiency with fewer titles, suggesting better resource utilization and cost-effectiveness.

This highlights that success and efficiency are related but distinct dimensions. While high spending can drive short-term success, long-term dominance depends on achieving a balance between performance and efficient use of resources.

⚙️ Feature Engineering

To enhance the analysis, additional features were created to better capture performance and market dynamics:

- **Revenue Growth %** → Measures league growth over time
- **Team Efficiency** → Win Rate / Team Value (captures cost-effectiveness)
- **Total Runs** → Combined score of both teams (match intensity)

New Feature: Match Competitiveness

- Defined as the absolute difference between team scores
- Helps understand how close or one-sided matches are

```
In [42]: # Create categories
def categorize_match(diff):
    if diff <= 10:
        return "Close Match"
    elif diff <= 30:
        return "Moderate Match"
    else:
        return "One-Sided Match"

matches['Match_Competitiveness'] = abs(matches['Team1_Score'] - matches['Team2_Score'])
matches['Match_Type'] = matches['Match_Competitiveness'].apply(categorize_match)

# Percentage distribution
match_percent = matches['Match_Type'].value_counts(normalize=True) * 100

# Plot
plt.figure(figsize=(8,5))

colors = ['#00C49F', '#FFBB28', '#FF8042'] # green, yellow, orange

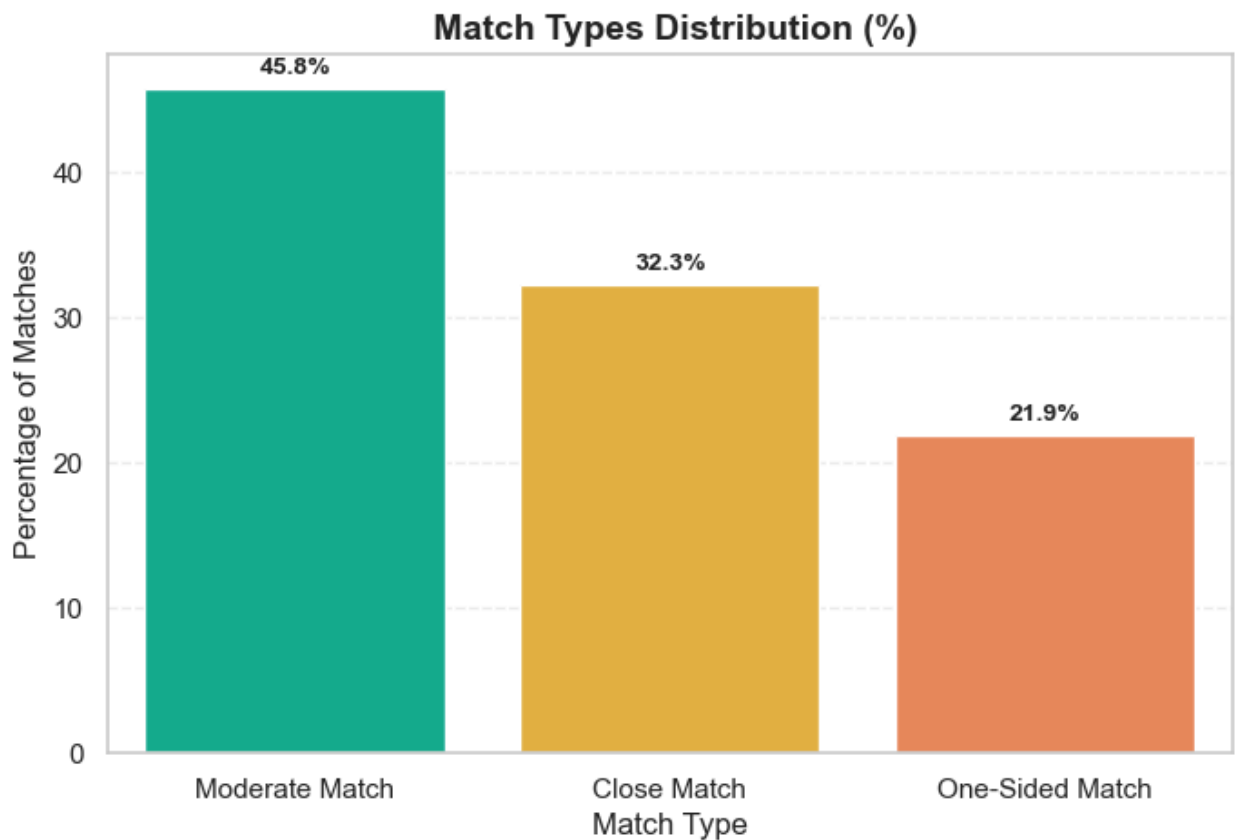
ax = sns.barplot(
    x=match_percent.index,
    y=match_percent.values,
    palette=colors
)

# Add % labels
for i, value in enumerate(match_percent.values):
    ax.text(i, value + 1, f"{value:.1f}%", ha='center', fontsize=10, fontweight='bold')

plt.title(" Match Types Distribution (%)", fontsize=14, fontweight='bold')
plt.xlabel("Match Type")
plt.ylabel("Percentage of Matches")

plt.grid(axis='y', linestyle='--', alpha=0.3)

plt.show()
```



🔥 “Do close matches attract more audience?”

```
In [43]: # Average attendance per match type
avg_attendance = matches.groupby('Match_Type')['Attendance'].mean()

plt.figure(figsize=(8,5))

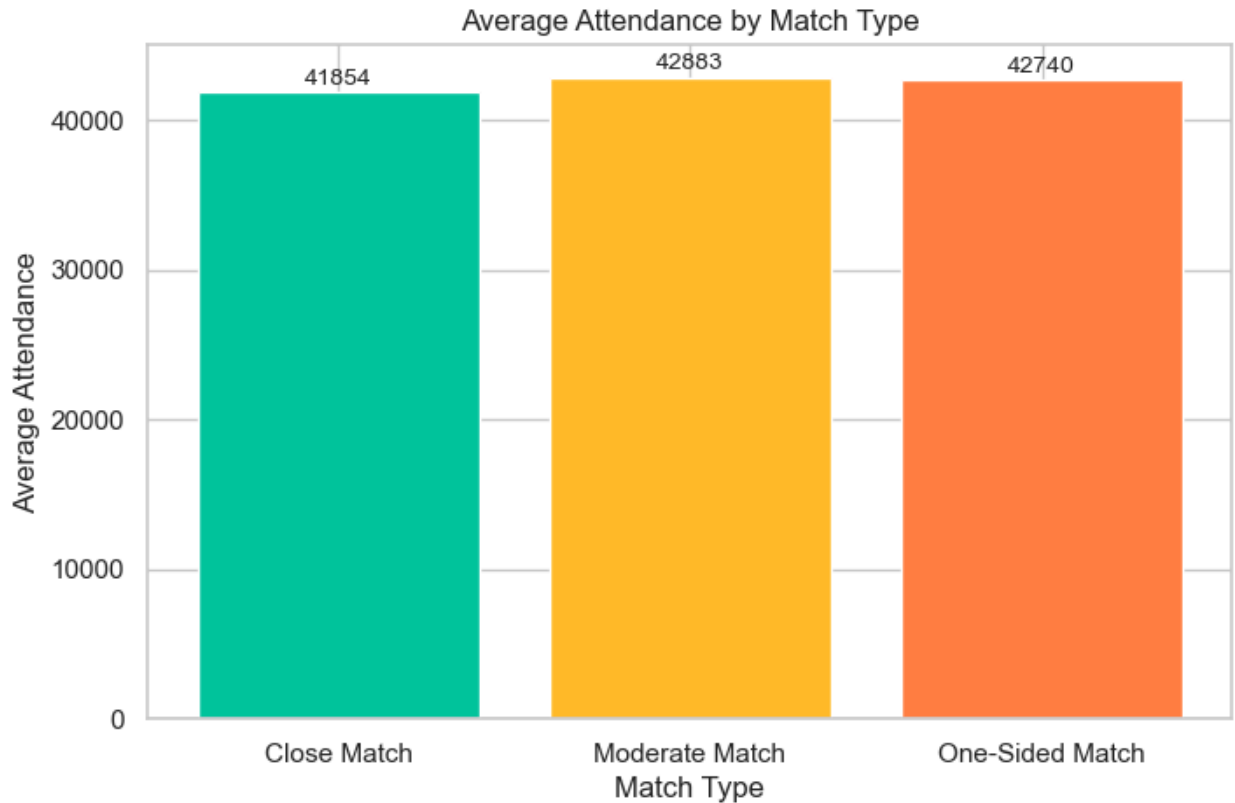
colors = ['#00C49F', '#FFBB28', '#FF8042']

plt.bar(
    avg_attendance.index,
    avg_attendance.values,
    color=colors
)

# Add values
for i, value in enumerate(avg_attendance.values):
    plt.text(i, value + 500, f"{int(value)}", ha='center', fontsize=10)

plt.title("Average Attendance by Match Type")
plt.xlabel("Match Type")
plt.ylabel("Average Attendance")
```

```
plt.show()
```



Insight

The average attendance across match types is very similar, with only minor differences between close, moderate, and one-sided matches.

This suggests that match competitiveness does not significantly influence attendance.

Attendance is likely driven more by factors such as team popularity, venue, and scheduling rather than how close the match turns out to be.

Which teams bring more audience?

```
In [48]: import numpy as np

team_attendance = matches.groupby('Team1')['Attendance'].mean().sort_values(ascending=False)
top_teams = team_attendance.head(10).reset_index()

plt.figure(figsize=(10,6))

# Proper gradient colors
```

```

colors = plt.cm.plasma(np.linspace(0, 1, len(top_teams)))

bars = plt.barh(
    top_teams['Team1'],
    top_teams['Attendance'],
    color=colors
)

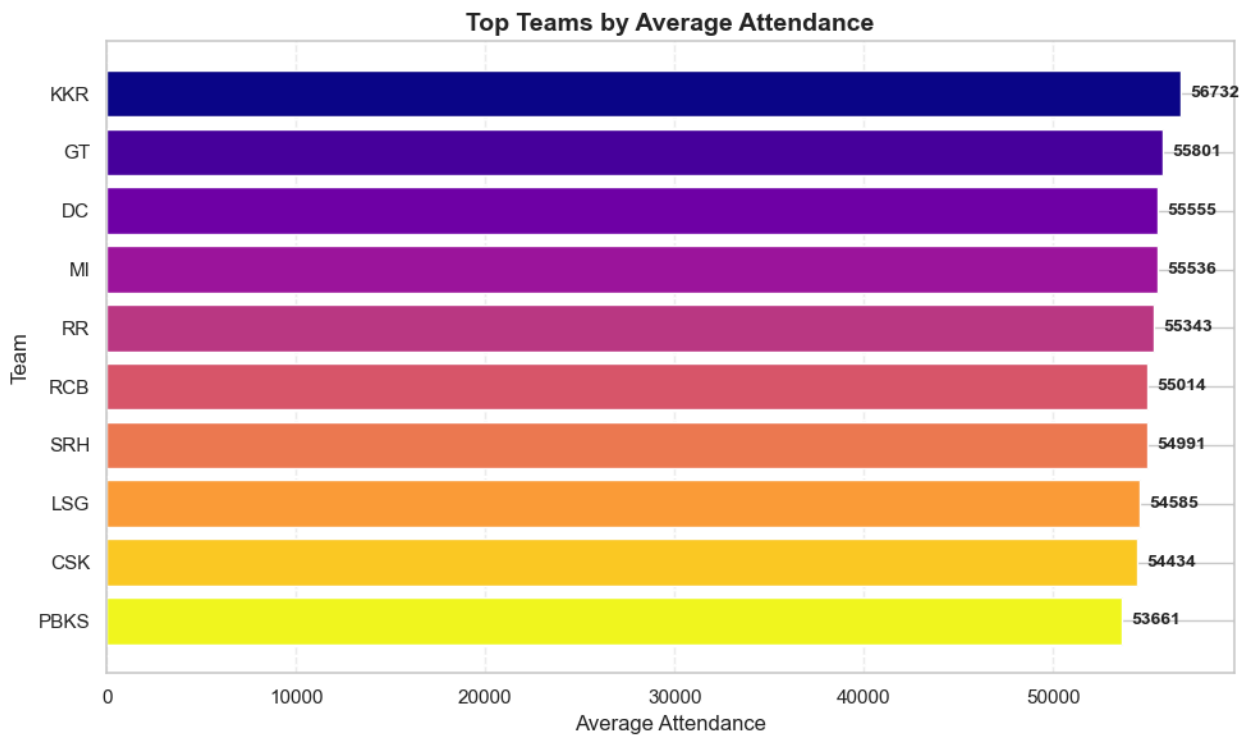
# Add values
for i, value in enumerate(top_teams['Attendance']):
    plt.text(
        value + 500,
        i,
        f"{int(value)}",
        va='center',
        fontsize=10,
        fontweight='bold'
    )

plt.title("Top Teams by Average Attendance", fontsize=14, fontweight='bold')
plt.xlabel("Average Attendance")
plt.ylabel("Team")

plt.gca().invert_yaxis()
plt.grid(axis='x', linestyle='--', alpha=0.3)

plt.tight_layout()
plt.show()

```



Stakeholder Insight: Team Popularity & Audience Engagement

The analysis shows a clear variation in average attendance across teams, indicating that certain teams consistently attract larger audiences.

Key Observations

- A few teams dominate in terms of attendance, suggesting strong brand value and fan following.
- Lower-performing teams attract comparatively fewer spectators, indicating weaker market presence.

Strategic Implications

For High-Attendance Teams:

- Leverage strong fan base for premium ticket pricing and sponsorship opportunities.
- Focus on maintaining performance and star players to sustain audience interest.

For Low-Attendance Teams:

- Invest in brand building and fan engagement strategies.
- Improve on-field performance to increase visibility and interest.
- Target local market activation and promotions to boost stadium turnout.

Business Insight

Attendance is driven more by team popularity and brand value than by match outcomes alone. Teams with strong identity and consistent performance are better positioned to generate higher engagement and revenue.

Limitations

- Player performance data (runs, wickets) is not available, limiting performance-based evaluation.
- Attendance data does not capture external factors such as weather, scheduling, or ticket pricing.
- Analysis is based on aggregated metrics, which may hide match-level variations.

Key Takeaways

- Team popularity is the strongest driver of attendance.
- Match competitiveness has minimal impact on audience turnout.
- Efficient teams are not always the most successful, indicating different strategic approaches.
- Spending strategy varies significantly across teams and leagues.

Final Conclusion

This analysis explored the growth, performance, and engagement dynamics across major T20 leagues, focusing on revenue trends, match characteristics, team strategies, and player investments.

Key findings reveal that league growth has been strong, driven by increasing revenues and expanding audience interest. However, match-level factors such as competitiveness and scoring have limited influence on attendance, indicating that fan engagement is not solely driven by on-field outcomes.

Instead, team-related factors play a more significant role. Teams with strong brand value, consistent performance, and established fan bases attract higher attendance and engagement. Additionally, the analysis shows that success and efficiency are not always aligned, as some teams achieve higher titles through greater investment, while others demonstrate better resource utilization with fewer trophies.

From a strategic perspective, long-term success in these leagues requires a balance between financial investment, team composition, and efficient resource management. Teams that combine strong performance with smart spending and brand building are better positioned to achieve sustained dominance.

Overall, this study highlights that in modern T20 leagues, success extends beyond the game itself — it is shaped by a combination of performance, strategy, and market-driven factors.

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